



Investigating the Unsteady Loads that Occur on the Surface of a BB2 Submarine Sail

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ABSTRACT

This report investigates steady and unsteady hydrodynamic loads on the BB2 submarine's sail using Computational Fluid Dynamics (CFD). A preliminary simulation was run on a 2-D cylinder to confirm the transient solver could recover canonical vortex shedding after which a NACA0022 section was adopted for the BB2 sailplane. A URANS $k-\omega$ SST framework in ANSYS Fluent was used with $y^+ < 1$ wall resolution and a domain extending 7.5c upstream and 15c downstream. The mesh-convergence study reduced velocity error from 7.5% to 0.4% while clarifying the cost-accuracy trade-off. Initial transient runs across angles of attack were between 0° - 18° and initial velocity of 6 - 24 m s^{-1} remained steady, showing no vortex shedding; after mesh smoothing, near-wall refinement, and solver updates, unsteadiness and vortex shedding development emerged at higher angle of attack of 22° and initial velocity of 24 m s^{-1} . Using the Strouhal number for the NACA0022 this number helped find the frequency for the simulation at different angles of attack and velocity. Results identified operating conditions that caused unsteady loads to become apparent, while outlining a path towards future research with less constrained budgets.

1. Introduction

The manoeuvring performance of a submarine is a critical parameter governing its operational effectiveness and safety. During dynamic manoeuvres, the vessel must maintain a controlled, predictable response to steering inputs to preserve stability and directional accuracy. Effective manoeuvrability relies primarily on the hydrodynamic performance of the sail assembly, which supports the vertical and horizontal control surfaces that generate the forces and moments required to alter the submarine's trajectory. These appendages manipulate the surrounding flow field to achieve controlled pitch and yaw responses, thereby maintaining the vessel's intended course.

However, at high angles of attack (AoA), the flow around the sail can become highly separated, leading to a substantial reduction in the effectiveness of these control surfaces. Once flow detachment occurs, the resultant loss of lift and control authority can induce unpredictable or unstable behaviour, severely compromising manoeuvrability. In extreme cases, this may leave the submarine momentarily uncontrollable, increasing the risk of collision, surface detection (for military applications), or loss of vessel stability. Understanding and predicting the onset of such flow separation phenomena is therefore

essential for ensuring the safe and reliable operation of submarines under high-load manoeuvring conditions.

Computational Fluid Dynamics (CFD) have long been used to calculate, verify, and model fluid dynamics around submarine sails [1]. With the advancement of high-performance computing, the shipbuilding industry has applied CFD to optimise the hull form design for ships and submarines [2]. This has led to remarkable developments in maritime design, and submarine sails have benefited significantly from investigations into constant-Reynolds-number experiments, such as those conducted by S.-K. Lee et al. [3] showed the development and eventual overtaking of turbulent flow around the sail as well as the creation of vortices behind the submarine as it progressed through the water. 3D experiments, such as those conducted by A. Rocca et al. [4], modelled the effects of 3D simulations on eddy currents and flow separation, enabling an accurate description of complex vortical structures and unsteady turbulent fluctuations surrounding the BB2 submarine geometry. Another experiment conducted by R. Broglia et al. [5] investigated the results of Reynolds-averaged Navier-Stokes (RANS) simulations on a towed, fully appended notional submarine model. While current studies provide valuable insights into vortex formation, turbulent wake structures, and hydro-acoustic emissions around submarines, there remains a notable lack

of focus on the unsteady loading characteristics of the BB2 sailplane across varying Reynolds numbers.

The use of different CFD models is also an essential distinction between investigations. The solving methods used by computers can be chosen depending on the complexity and computational demand of the issue at hand. Models range from RANS methods, which average flow across discrete boundaries and do not capture turbulent boundary layers well, to DNS methods, which fully resolve flow characteristics. The computational costs of DNS and other high-resolution methods are often both unfeasible and unnecessary for investigations where high turbulence and small boundary-layer interactions are not in focus, as the individual runtimes can approach multiple months, especially without access to supercomputers. Methods that sit in this range of specificity include high-fidelity Large Eddy Simulations (LES), where large computational grids and excellent spatial resolution are key to their success [5]. One of the first computations of this type came from Posa and Balaris [6], where a submarine was analysed within a constant Reynolds number. Wang et al. [7] continued such LES experiments, albeit with a lower Reynolds number. LES is still expensive, and so wall-modelled LES (WMLES) or hybrid RANS/LES methodologies (such as Detached Eddy Simulation, DES) were also exploited [5]. Studies comparing LES, DES, and RANS predictions done by Alin et al. [8], Bhushan et al. [9], Chase and Carrica [10], Chase et al. [11], and ChangSeop et al. [12] found unsteady RANS (URANS) computations were a valid, affordable alternative to WMLES or DES, especially for handling high Reynolds numbers.

The development of empirical variables, such as the Strouhal Number (St), has progressed rapidly recently. The relationship between St and the Reynolds number was investigated by Jiang and Cheng [13] using DNS at low Reynolds numbers and found that the St values for 2D and 3D flows agree well in the laminar regime up to approximately 270. They also found, however, that the wake flow becomes increasingly turbulent with further increases in Re , and this causes the St value for the 3D flow to gradually deviate from its 2D counterpart, since for turbulent 3D flows the vortex shedding frequency scales with a length smaller than the wake width. Shojaeefard et al. [14] conducted a RANS study of St effects on the stability derivatives of an autonomous underwater vehicle, and found St to be increasingly important at higher Reynolds numbers, and indicated the

considerable dependencies of all stability derivatives on St . An interesting investigation published by Cohen and Cleary [15] analysed the Strouhal values of marine animals and proposed that the St values produced by these animals produce optimal vortex formation. Calculation methods for St values have also undergone various improvements, with Chang K. and Dae K. [16] proposing a new method to determine the St of a rectangular cylinder in uniform flow.

Mesh strategy development is an essential part of successful CFD analysis and the capture of valid results. An assessment of hydrodynamic characteristics and computational resources done by Abidin et al. [17] analysed the impact of various mesh sizes on results. They employed a RANS turbulence model with $k-\omega$ SST to analyse the resistances of the MARIN's BB2-class submarine upon various mesh densities from 1 to 97 million cells. The findings showed that the results were validated with errors ranging from 0.3% to 10% across different mesh densities. Furthermore, using a standard PC with the OpenFOAM meshing tool yielded acceptable accuracy, with less than 5% error at lower mesh densities. The results showed that the importance of mesh size plateaued at 4 million cells with their specific design, and further refinements in size were marginal in improving accuracy. A study conducted by Enkai et al. [18] using a similar method with RANS equations and the SST $k-\omega$ turbulence model found that a coarser mesh of 5 million cells gave only a ~5% error compared to a finer mesh of 26 million cells. ChangSeop et al. [12], using the URANS solver from STAR-CCM+ ver. 16.06R8 (Siemens, 2021) found that using a medium-grain mesh of 6.5 million cells yielded an impressive 0.6% error, while a coarse mesh yielded less than 5%. These studies lend credence to the validity of coarse meshes in preliminary and early concept design, with all reporting a maximum 5% error with the coarse-mesh method. More valuable data, however, must come from medium and fine meshes.

2. Experimental Approach

The BB2 sail configuration was established to represent the aerodynamic conditions acting on the structure accurately. The airfoil profile selected for the analysis was a standard section developed by the National Advisory Committee for Aeronautics (NACA).

2.1. The BB2 Submarine Sail

To simplify the computational study a symmetrical airfoil section, the NACA 0022, has been selected as the primary geometry. Symmetrical airfoils (i.e., those with identical upper and lower surface shapes and no camber) offer several advantages in computational fluid dynamics (CFD) investigations: they present a zero lift condition at zero angle of attack, they simplify the behaviour of the mean line and pressure distribution (thereby reducing complexity in the solver setup), and they allow clearer isolation of angle of attack, Reynolds number, and wake-dynamics effects (see e.g., the review of NACA airfoil geometry generation and behaviour in the four-digit series) [16]. Moreover, using a well-documented classical airfoil such as NACA 0022 aids in validating computational setups by comparing results with prior numerical/experimental data (for example, recent work on NACA 0022 steady and dynamic tests) [17].

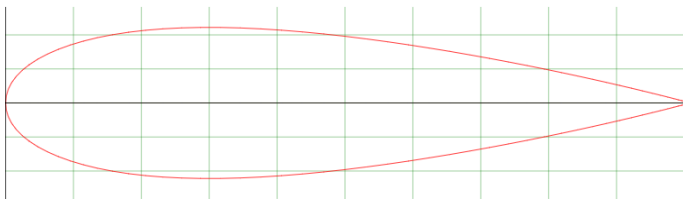


Figure 1 - NACA0022 Airfoil

2.2. Ansys Fluent

ANSYS Fluent is a commercial computational fluid dynamics (CFD) software package that employs the finite volume method to discretise and solve the governing fluid-motion equations (Navier–Stokes) over complex geometries and flow domains [18]. It supports a wide range of physics and is an industry leader in the modelling and designing of fluid-based components and vehicles [19]. Fluent provides an accessible yet highly sophisticated environment necessary for technical investigations into fluid dynamics across time, as is the topic of this report, and was instrumental in simulating both steady and unsteady flow regimes, enabling the detailed investigation of vortex formation, wake dynamics, and the influence of geometric and flow parameters on aerodynamic performance.

2.3 Initial Unsteady Test with Cylinder

Before commencing the main airfoil simulations, a preliminary series of tests was conducted on a two-dimensional circular cylinder to evaluate the solver setup, identify suitable mesh parameters, and observe the onset

of unsteady flow phenomena such as vortex shedding. The cylinder geometry was selected because it serves as a well-established benchmark in fluid dynamics, often used to validate computational models before applying them to more complex geometries [20], [21]. By reproducing this standard case, it could be confirmed that the transient solver in ANSYS Fluent would accurately capture periodic wake behaviour and to determine an appropriate balance between computational cost and numerical accuracy.

The initial setup involved defining a uniform flow domain around the cylinder with sufficient spatial extent to avoid blockage and boundary interference, consistent with the recommendations of bluff-body CFD studies [22], [23]. The Reynolds number was selected to fall within the subcritical regime ($Re \approx 10^4$ – 10^5), where laminar vortex shedding typically occurs, allowing the formation of a von Kármán vortex street in the wake region [24], [25]. Boundary conditions included a uniform-velocity inlet, a static-pressure outlet, symmetry at the upper and lower boundaries, and a no-slip condition at the cylinder wall. The computational mesh was generated using an unstructured triangular mesh, with radial refinement near the cylinder surface to capture boundary-layer separation accurately. A grid-independence study was performed by varying the cell count from approximately 40,000 to 700,000 elements to establish mesh convergence, following best practices demonstrated in previous numerical investigations [26], [27].

After achieving steady-state convergence, the model was transitioned to a transient simulation to capture the time-dependent wake behaviour. Several time-step sizes were evaluated, progressively reduced until sufficient temporal resolution was obtained to resolve periodic oscillations in the lift and drag coefficients. The resulting velocity and pressure fields revealed the alternating vortex-shedding pattern characteristic of unsteady bluff-body flow. The Strouhal number (St) was calculated from the dominant frequency of lift fluctuations and compared with the established empirical relationship ($St \approx 0.21 \pm 0.02$) for subcritical Reynolds numbers [28], [20]. Agreement within this range validated both the transient solver setup and the model's overall numerical accuracy.

During these preliminary runs, challenges primarily related to solver stability and computational efficiency were encountered. Larger time steps tended to suppress unsteadiness, whereas minimal values significantly

increased computation time due to the limited four-core processing capacity available on ANSYS Fluent workstations. Despite these constraints, the solver configuration, turbulence model, and boundary conditions successfully reproduced the expected unsteady wake dynamics. Consequently, the cylinder simulations provided a validated baseline for transient modelling and directly informed the mesh refinement, time-step selection, and convergence strategies used in subsequent airfoil and sail analyses.

2.3 Geometry and Domain Setup

The geometry and computational domain of the simulation were established using widely adopted configurations from the literature, reflecting standard BB2 submarine geometry and flow analysis practices. The BB2 sailplane is characterised by a NACA 0022 airfoil profile, which provides a symmetric section suitable for controlled investigation of flow separation, vortex shedding, and unsteady wake behaviour. This profile was used directly to replicate the sail geometry in the numerical model, ensuring fidelity with previously validated studies.

The domain boundaries were sized according to established best practices to minimise artificial effects from boundary conditions and to allow the flow structures to develop downstream of the sail fully. The literature commonly recommends extending the downstream domain to at least 15 chord lengths beyond the airfoil, which is sufficient to capture the formation, evolution, and dissipation of wake vortices without numerical reflection. Similarly, an upstream distance of approximately 7.5 chord lengths ahead of the airfoil ensures that the incoming flow field remains uniform and undisturbed by any recirculation or boundary effects. Initial conditions require a distance to stabilise and become steady; therefore, a small distance between the mesh boundary and the airfoil would result in incomplete flow and, consequently, invalid results. This domain sizing strategy aligns closely with configurations used in previous numerical investigations of the BB2 geometry, thereby enhancing the reliability of the results and enabling direct comparison with published experimental and computational data.

2.4 Mesh Generation

Mesh generation is one of the most critical components in ensuring that a CFD simulation accurately captures the unsteady flow field. The mesh essentially defines how the computational domain is discretised, with each cell representing a finite control volume over which the governing Navier–Stokes equations are solved. Since CFD solvers compute flow properties at discrete spatial locations, the resolution and quality of these cells directly dictate the fidelity and stability of the solution. A coarser mesh, also known as low-fidelity discretisation, results in larger cell sizes and therefore lower spatial resolution. This can lead to the averaging of key flow features over large distances, which is particularly problematic in simulations involving flow separation, vortex shedding, or other inherently unsteady and turbulent behaviours.

In laminar or steady flows, where velocity and pressure gradients vary smoothly across the flow field, a low-resolution mesh may still yield reasonably accurate results. However, unsteady flows are characterised by complex vortex structures, shear-layer instabilities, and fluctuating turbulent eddies, all of which require a sufficiently fine mesh to resolve. If the mesh is too coarse, these turbulent phenomena are effectively "smoothed out" and not captured by the solver, resulting in a simulation that may be numerically stable but physically inaccurate. This leads to underprediction of key quantities such as Strouhal frequencies, lift fluctuations, wake structures, and pressure distributions.

Conversely, creating an excessively fine mesh can introduce a different set of challenges. While higher resolution improves accuracy, it also dramatically increases the total number of computational cells, thereby increasing solver time, memory usage, and computational cost. For transient, unsteady flow simulations, where small time steps are often required to satisfy numerical stability conditions, such as the Courant–Friedrichs–Lewy (CFL) criterion, this can result in simulations that take impractical amounts of time to complete, particularly in the absence of high-performance computing (HPC).

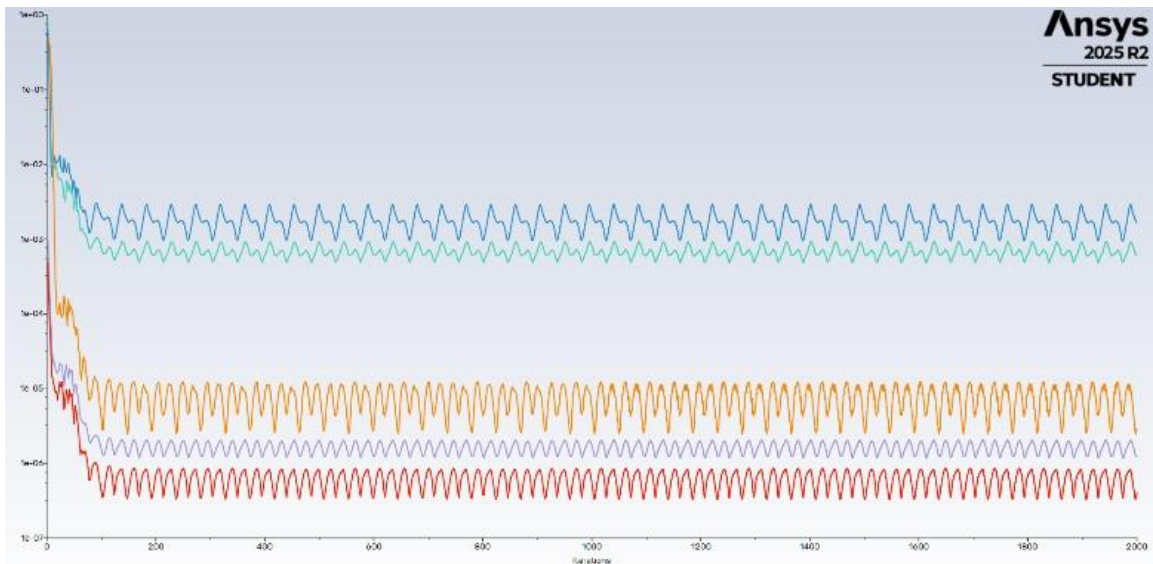


Figure 2 2 - Residuals show decline followed by cyclical behaviour, validating the experimental design

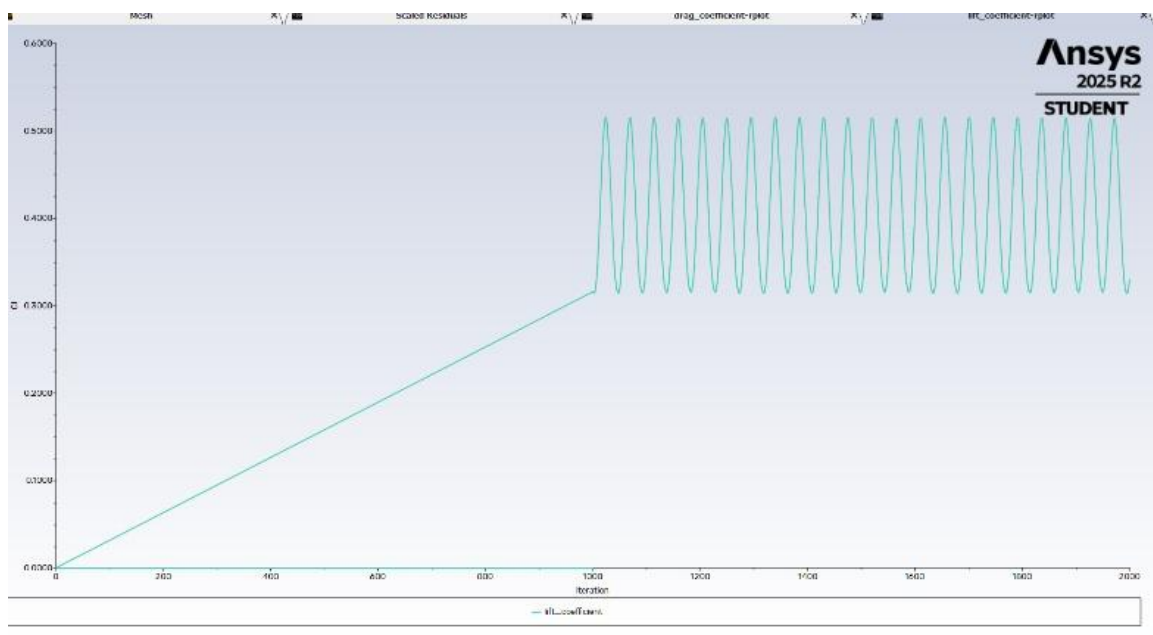


Figure 33 - C_1 shows sinusoidal behaviour following initial period of stabilisation

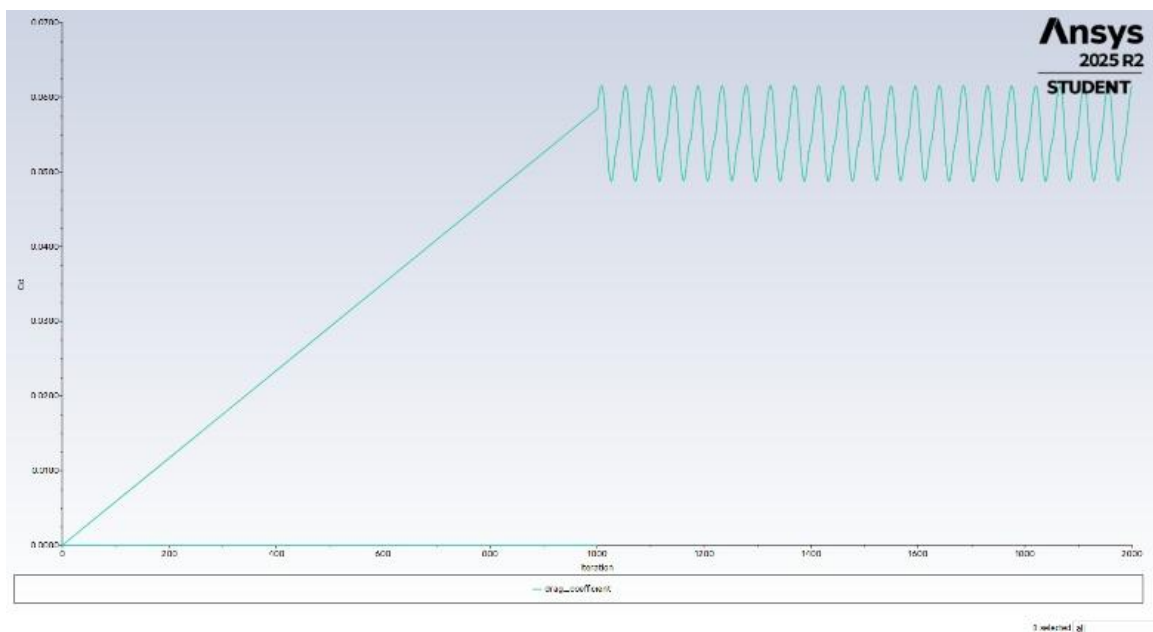


Figure 44 - C_d shows similar behaviour to C_1 , validating results

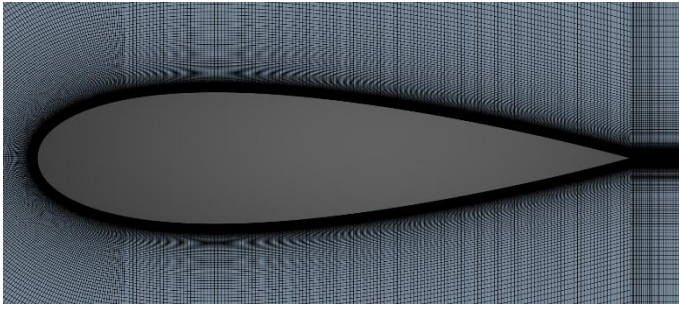


Figure 55 - Mesh Generated around airfoil

In this simulation, a final mesh consisting of approximately 700,000 cells was generated. This cell count was chosen to balance resolving the key unsteady flow structures with reasonable computational cost. The mesh density was concentrated around the sail and wake region to accurately capture flow separation, vortex shedding, and pressure fluctuations, while gradually coarsening toward the far-field boundaries to reduce the total cell count without compromising solution quality. A structured boundary-layer refinement was applied near the surface to ensure adequate resolution of near-wall flow behaviour, particularly important for correctly capturing the onset of separation and the development of shear layers. This resolution level is consistent with those used in similar studies. It has been shown to yield stable, accurate results for unsteady Reynolds-Averaged Navier–Stokes (URANS) simulations of comparable geometries. By adopting an intermediate mesh density, the simulation achieved good spatial fidelity while keeping the required computational resources and runtime within practical limits.

2.5 Final Parameter Design

Table 1 - Final Design Parameters

Parameter	Method
Model Used	k– ω
Time Formulation	Transient (Unsteady)
Time Step Size	0.001 s
# of Time Steps	4000
Max Iterations	20
Initial Velocity	6 ms ⁻¹
Pressure–Velocity Coupling	PISO

3. Results & Discussion

3.1 Mesh Convergence

To assess the impact of grid density on solution accuracy and computational cost, a mesh convergence study was conducted. After methodically refining the computational

mesh from 43,750 to 700,000 cells, the anticipated maximum velocity approached the analytical benchmark of 6.64 m/s. Numerical convergence and enhanced solution integrity were demonstrated by the corresponding error, which dropped from 0.075 to 0.004. The trade-off between precision and computing efficiency was evident, though, as the improvement also led to a significant increase in wall-clock time.

Table 2 - Mesh Sizing

Mesh Size	Wall Clock Time (s)	Predicted Max Velocity (m/s)	Analytical Max Velocity (m/s)	Error
43750	74.19069	6.6449779	6.64	0.075
175000	81.828	6.6431456	6.64	0.047
393750	134.809	6.6404423	6.64	0.007
700000	901.42852	6.6402764	6.64	0.004

3.2 Mesh

To minimise boundary effects and guarantee a fully developed flow around the shape, the computational domain was designed. To ensure proper flow growth and wake dissipation, the domain is composed of an extended rectangular outlet region and a semi-circular intake region, as seen in Figure 7. To precisely capture boundary-layer gradients and vortex generation, a structured mesh was created utilising hexahedral cells, with refined cell density along the wake region and close to the geometry. To reduce computational costs while preserving numerical stability, gradual mesh coarsening was applied at the outer edges. Accurate flow prediction was supported by the mesh topology's smooth cell transition and orthogonality, which prevented skewness-related errors and artificial diffusion.

3.3 Calculation of y^+

To ensure sufficient near-wall mesh resolution and precise capture of boundary-layer phenomena, the dimensionless wall distance, y^+ , was evaluated. The goal of this study was to place the first cell centroid within the viscous sublayer, where the velocity profile stays linear, by keeping y^+ values below 1. This removed the requirement for wall-function approximations by guaranteeing that the wall-adjacent cells directly resolved the viscous effects. Accurately forecasting wall shear stress, boundary-layer formation, the beginning of flow separation, and vortex formation all depends on this resolution. The obtained y^+ distribution ensured the accuracy of the calculated wall-bounded flow characteristics by confirming that the mesh was sufficiently fine in the near-wall regions.

3.4 Steady State Solution

Before transient analysis, a converged flow field was first established by a steady-state simulation. This method ensured that the initial conditions for the time-dependent simulation were both numerically stable and physically consistent, while also allowing residual transients to dissipate. To resolve boundary-layer flows and flow separation under unfavourable pressure gradients, the shear-stress transport (SST) $k-\omega$ turbulence model was used. The SST formulation combines the benefits of the $k-\epsilon$ model in the far field, which reduces sensitivity to freestream turbulence levels, with the classic $k-\omega$ model near the wall, where it accurately predicts viscous sublayer behaviour.

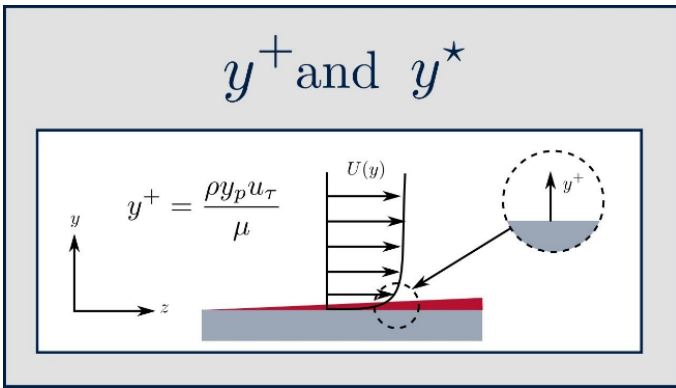


Figure 66 - y^+ calculation method alongside visual diagram

To evaluate unsteady vortex dynamics in later transient simulations, the steady-state solution served as a baseline, providing information on the mean flow characteristics, pressure distribution, and wake shape. Key flow variables, such as lift, drag, and velocity magnitude, were stabilised, and residual reduction was used to monitor convergence. It was confirmed that the mesh and turbulence model combination was sufficient to capture steady aerodynamic features before moving on to the unsteady phase, based on the solution's consistent flow behaviour and minimal residual oscillations. Even before the transient analysis began, unstable aerodynamic behaviour was already being recorded, as further evidenced by the steady-state results shown in Figures 2-4. Both the lift force and the pressure force showed an apparent change from a linear growth to a periodic oscillation pattern as the number of rounds rose. This oscillatory behaviour suggests that, even when numerically confined to a steady formulation, the flow itself is physically unsteady because the steady solver begins to resolve the inherent vortex shedding in the wake of the body. After roughly 900–1000 cycles, the signal

stabilises into a repeating waveform in the pressure and lift coefficient plots, indicating the formation of a self-sustaining vortex street. Likewise, the residual history shows that after an initial decline, the variables under observation bounce about a mean level rather than converging to a single value. This behaviour confirms that the chosen mesh and turbulence model can resolve time-dependent flow instabilities and is typical of bluff-body or high-AoA flows where periodic shedding occurs. As a result, these steady-state oscillations demonstrate that unstable stresses are acting on the geometry, confirming the need for a transient method to capture the flow's temporal history accurately. The findings show that even the stable solver can identify the onset of dynamic wake structures, such as alternating vortices, and provide a trustworthy mean flow field that can be used to start the transient simulation.

3.5 Transient Solution

Switching from steady-state to transient flow simulations brought a noticeable increase in computational complexity and solver time. While the steady-state cases converged quickly and were relatively simple to set up, the transient runs took considerably longer to develop and stabilise. This was expected, as transient simulations capture time-dependent changes in the flow field, such as vortex formation and shedding, which require smaller time steps, tighter convergence criteria, and much greater computational effort to resolve accurately.

One significant challenge throughout this process was the limited availability of computing resources. All simulations were performed in ANSYS Fluent, which limited each workstation to 4 CPU cores. This significantly limited the level of parallel processing achievable and, in turn, increased the total simulation time. Transient flow simulations are inherently demanding because they must resolve many time steps for even a short duration of physical time, and each time step requires several iterations to ensure numerical stability. As a result, even a few seconds of real-time flow could take many hours to compute.

The study tested a range of flow conditions, varying both angle of attack (0° , 5° , 10° , 15° , and 18°) and velocity (6–24 m/s, in 2 m/s increments). A middle-range case was first used to confirm that the transient solver setup was functioning correctly. Interestingly, the flow remained completely steady under these conditions, showing no

evidence of vortex shedding or unsteady wake behaviour. This suggested that either the Reynolds number was below the threshold for periodic vortex formation or that further refinement in the mesh and time-step resolution was needed to capture unsteady effects accurately.

In the early stages of testing, a coarser mesh and simplified solver settings were used to reduce computation time and validate the overall setup. Under these conditions, each simulation took roughly one to three hours to complete. This relatively quick turnaround was mainly due to the larger mesh elements, the larger time step, and the lower maximum iteration limit per step. However, this came at the cost of accuracy as the coarse mesh and simplified settings could not fully capture smaller flow features or transient effects.

As the simulations progressed across the tested range of angles of attack and inflow velocities, a consistent pattern emerged: none of the examined cases showed evidence of vortex formation or periodic shedding from the airfoil surface. Each configuration, starting at 0° and incrementally increasing to 18° , produced flow fields that remained steady throughout the simulation. The absence of unsteady wake development or fluctuating lift forces suggested that the flow over the NACA 0022 airfoil under the tested conditions remained fully attached and laminar. Furthermore, the transient results showed negligible deviation from their steady-state counterparts, indicating that, within the investigated Reynolds number range, the onset of vortex shedding or other unsteady flow phenomena did not occur.

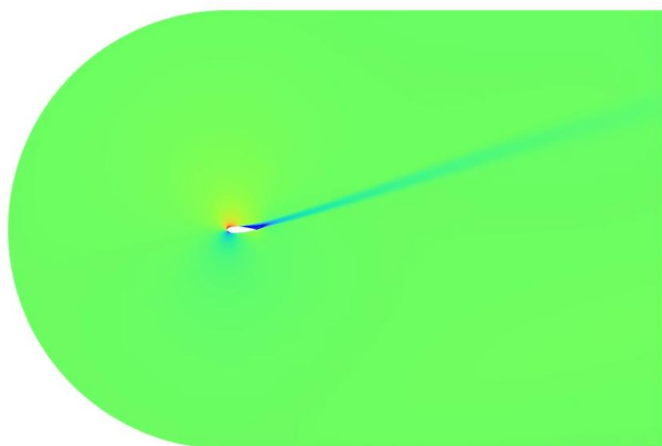


Figure 7 - Velocity Contour (24m/s at 18° AoA)

From this stage onward, it became evident that significant modifications were required to improve both the mesh quality and the solver configuration used in the

simulations. The initial results indicated that the existing setup lacked sufficient spatial resolution and smoothness to capture boundary-layer behaviour and near-wall flow features accurately. Consequently, the first refinement step involved generating a denser, more uniformly graded mesh, with particular emphasis on regions of high velocity and pressure gradients around the airfoil surface.

During the preliminary simulations, one of the critical issues identified was a discontinuity in the meshed sections, specifically a visible interface between the leading-edge region of the airfoil and the adjoining mesh zone. This geometric segmentation produced an artificial separation line, introducing spurious numerical oscillations and localised unsteady flow behaviour within the boundary layer. Such discontinuities can distort the flow field development, compromise solution stability, and lead to non-physical fluctuations in aerodynamic coefficients. To mitigate these effects, the mesh was reconstructed to ensure smooth transitions between domains and improved orthogonality and aspect ratios near the surface. These refinements were essential to achieve a more accurate representation of the boundary-layer evolution and to enhance the overall reliability of the transient flow predictions.

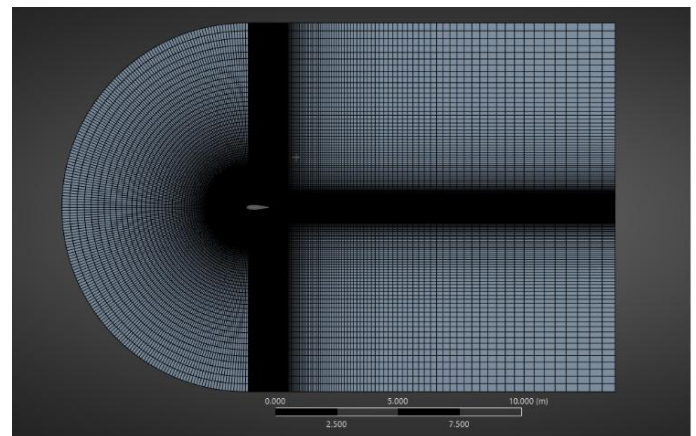


Figure 8 - New mesh closer and with more cells

Several targeted improvements were made to the mesh topology and near-wall resolution to enhance simulation accuracy and numerical stability. A constant, progressive change in element size was introduced throughout the domain by carefully smoothing the interfaces between mesh sections of varying density. The sudden discontinuities observed near the airfoil's leading edge, which had caused artificial flow disturbances and non-physical variations in the boundary layer, were removed by this improvement. The final mesh had a very consistent

element-size gradient, which made transitions between sections seamless and improved the overall quality of the captured flow field.

In the near-wall regions, typical element sizes were refined to about 1 mm, and the global mesh resolution was significantly enhanced. Improved resolution of pressure and velocity gradients around important aerodynamic structures, including the leading and trailing edges, was enabled by this increased spatial density. Furthermore, to better depict the viscous sublayer and near-wall shear behaviour, the first layer height in the inflation zone was reduced to 6.2844×10^{-6} m. This modification ensured that the wall-bounded flow was resolved directly instead of modelling it through wall functions by producing y^+ values that fell within the desired range for low-Reynolds-number turbulence modelling ($y^+ < 1$). The prediction of shear stress distribution and boundary-layer growth along the airfoil surface was greatly enhanced by reaching this low y^+ domain.

A two-dimensional mesh configuration was maintained throughout to balance computational efficiency with physical accuracy. The combination of fine near-wall resolution, smooth mesh transitions, and well-controlled y^+ values produced a more stable numerical environment and a physically consistent representation of the flow. Collectively, these refinements established a robust computational foundation for subsequent transient analyses, reducing numerical artefacts and improving solver convergence.

Following the mesh refinement, the solver configuration was systematically reviewed and adjusted to ensure the simulation environment was optimised to capture unsteady flow behaviour. Each setting was examined in detail to establish a configuration that would provide numerical stability while preserving the temporal fidelity required to resolve transient aerodynamic phenomena.

Initially, the turbulence model employed was the standard $k-\omega$ (two-equation) model, which was subsequently replaced with the $k-\varepsilon$ model during early testing. However, after consultation with academic supervisors and peers, the decision was made to revert to the $k-\omega$ formulation, as it demonstrated superior performance in resolving near-wall flow structures and unsteady wake dynamics. Among the modifications applied within the turbulence model, the curvature correction option was explicitly disabled. This correction, while beneficial for

cases involving pronounced streamline curvature or rotational effects, was found to interfere with the accurate prediction of wake development and fluctuating aerodynamic loads in the present configuration. Deactivating this feature improved the consistency of the unsteady load predictions and the stability of the vortex-shedding behaviour in the wake.

A key practical adjustment concerned the working fluid used in the simulations. Although the initial setup utilised water to match the experimental test conditions, computational limitations quickly became evident. The solver produced repeated floating-point errors and rapid divergence under water-based flow conditions, due to the high density and viscosity contrasts. To maintain solver stability and achieve convergence, the fluid medium was therefore changed to air, which provided a lower-density flow field and significantly reduced numerical stiffness. While this substitution introduced some scaling differences, it allowed the unsteady flow characteristics to be effectively captured within the computational limits of the available hardware.

The inlet boundary conditions were also refined to reproduce realistic turbulence characteristics better. The turbulence intensity was set to 1%, representing a moderately low turbulence inflow suitable for controlled aerodynamic analysis, while the turbulence length scale was defined as the airfoil chord length (1 m). This configuration ensured that the inlet turbulence parameters remained physically consistent with the simulation's geometry and flow regime.

Finally, the transient simulation parameters were optimised to balance temporal resolution and computational cost. A total of 4000 time steps were employed, each with a time-step size of 0.001 s, providing sufficient temporal granularity to resolve unsteady vortex dynamics. The maximum number of iterations per time step was set to 20, allowing adequate residual convergence even if the flow field stabilised earlier within a given step. This configuration provided stable, efficient solver performance while maintaining high-resolution tracking of unsteady aerodynamic phenomena across all simulated cases.

With the revised solver and mesh configuration, the test matrix was further extended to include a higher angle of attack (22°) and a higher freestream velocity (24 m/s). Under these more aggressive flow conditions, the

simulation finally exhibited clear evidence of unsteady flow development in the wake region. Periodic fluctuations in the pressure and velocity fields began to emerge downstream of the airfoil, indicating the onset of vortex shedding and confirming that the refined numerical setup could resolve the transient flow phenomena that had previously remained undetected.



Figure 9 - Velocity Contour (12m/s at 22° AoA)

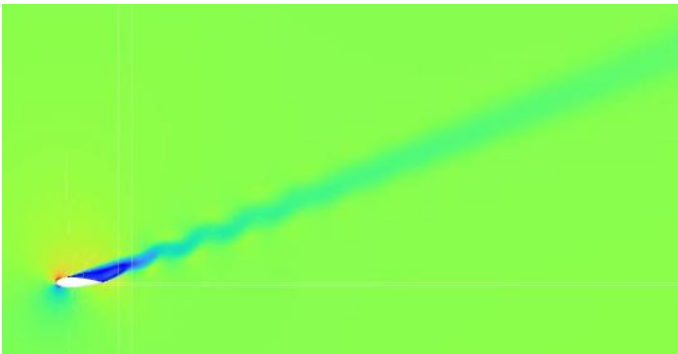


Figure 10 - Velocity Contour (24m/s at 22° AoA)

3.6 Strouhal Number

To estimate the dominant unsteady loading frequency acting on the BB2 sail section, the Strouhal number is applied based on representative data for a NACA 0022 airfoil. The Strouhal number characterises vortex-shedding behaviour and is defined as $St = \frac{fD}{U}$, where f is the shedding frequency, D is a characteristic length, and U is the free-stream velocity. Experimental studies show that symmetric airfoil sections such as the NACA 0022 exhibit relatively consistent Strouhal behaviour in the post-stall regime, with St typically ranging from 0.1–0.25 depending on angle of attack and Reynolds number. By selecting an appropriate characteristic length (e.g., sail chord or trailing-edge thickness) and known operating velocity for the BB2 case, this relationship enables a first-order prediction of the expected vortex-shedding

frequency to compare against CFD-derived unsteady load spectra.

4. Conclusion

This project developed and validated a computational fluid dynamic study to quantify loads on a BB2-class submarine sail and to identify the conditions under which unsteady forces create vortex shedding. The approach was verified using a transient cylinder benchmark that reproduced vortex shedding, while a mesh-convergence study was done to increase accuracy, as cell count increased from 44,000 cells to approximately 700,000. A URANS $k-\omega$ SST setup was used with a $y^+ < 1$ and carefully smoothed transitions captured boundary-layer behaviour without excessive computational cost. Tests were done across a range of angles of attack and velocities that showed flows at moderate AoA were largely steady for the simulated Reynolds range, but unsteady loads emerged at higher AoA and higher initial velocity, where vortex shedding structures formed and periodic load fluctuations appeared.

There are two applications for the findings. First, manoeuvring envelopes that force the sail to high AoA are more vulnerable to load oscillations, which can weaken directional stability and control-surface authority. For this reason, design margins and control-law tuning should pay special attention to these conditions. Second, URANS with high-quality near-wall resolution provide a cost-effective way to bracket frequencies using a Strouhal analysis and a map stable vs. unsettled regimes for early-stage design and scenario screening.

When more computation is available, future work should: (1) restore water properties and extend Reynolds numbers to BB2-representative ranges; (2) validate against targeted experiments for NACA0022/BB2 sections at high AoA; (3) investigate hybrid RANS–LES or DES/WMLES in localised regions to resolve wake dynamics and broadband content.

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